
10
YEARS
07.NOV.2024
Dialoguer avec vos documents
Découvrez la magie du RAG avec et sans OpenAI
ELEVATING THE DEVELOPERS' COMMUNITY
devday
GET INSPIRED
WORLDLINE

Raphaël Semeteys

Head of DevRel

Architecte Senior, Expert Open Source

- Atos puis Worldline depuis 1999
- Basé à Paris, France
- Professeur de Yoga
- raphiki.github.io



@RaphaelSemeteys

dev.to/raphiki/

linkedin.com/in/raphaelsemeteys/

Who are we?

We design payments technology that powers the growth of millions of businesses around the world.



7000+ engineers
in over 40 countries



Managing 43+ billion
transactions per year



€250M spent in R&D
every year



Handling 150+
payment methods

Cas d’usage : RPG Codex

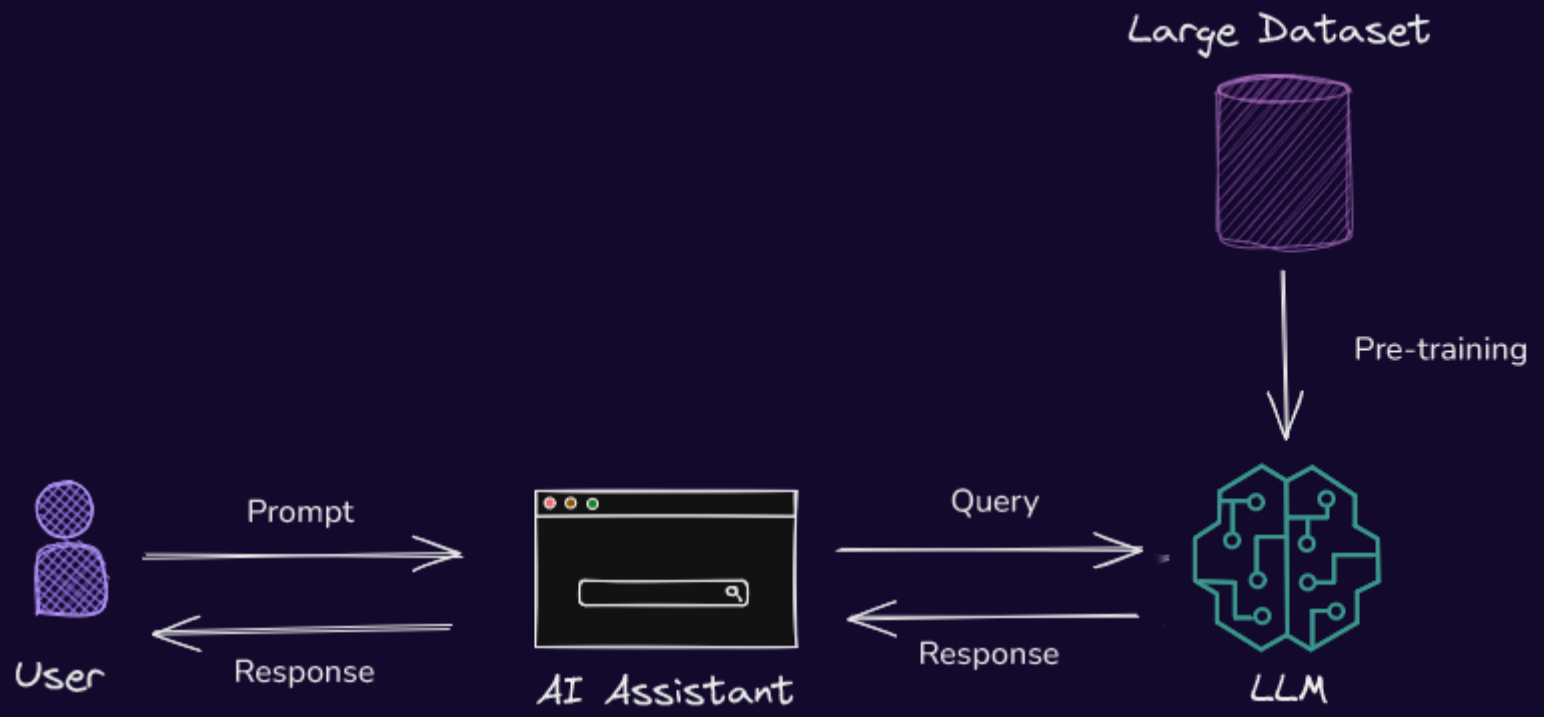
***Assistant** qui répond aux questions, clarifie les règles et fournit des informations sur des jeux de rôle en se basant sur leurs **documents** officiels*





https://raphiki.github.io/fr/garous.html

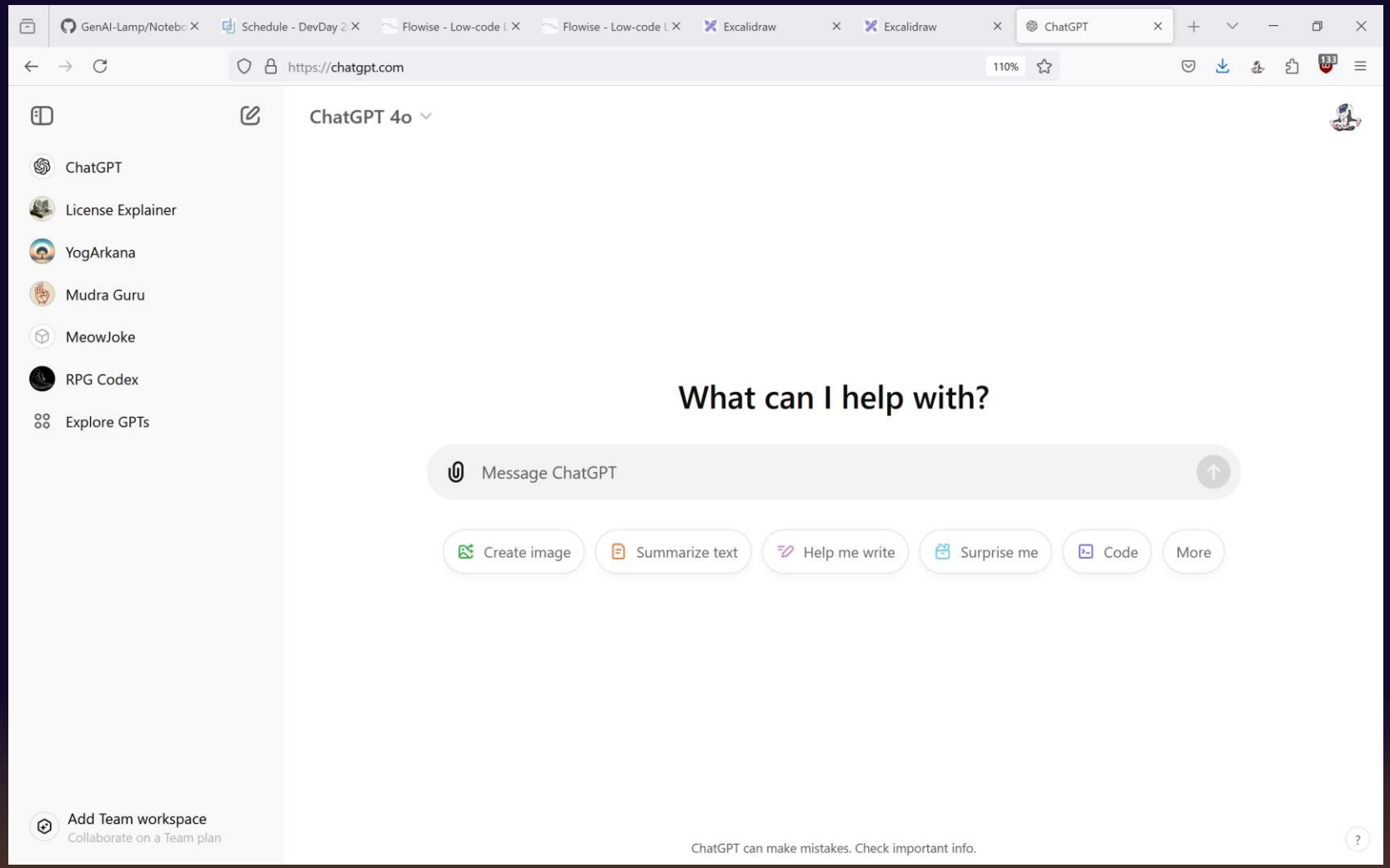
Chat Completion



Chat Completion

DEMO

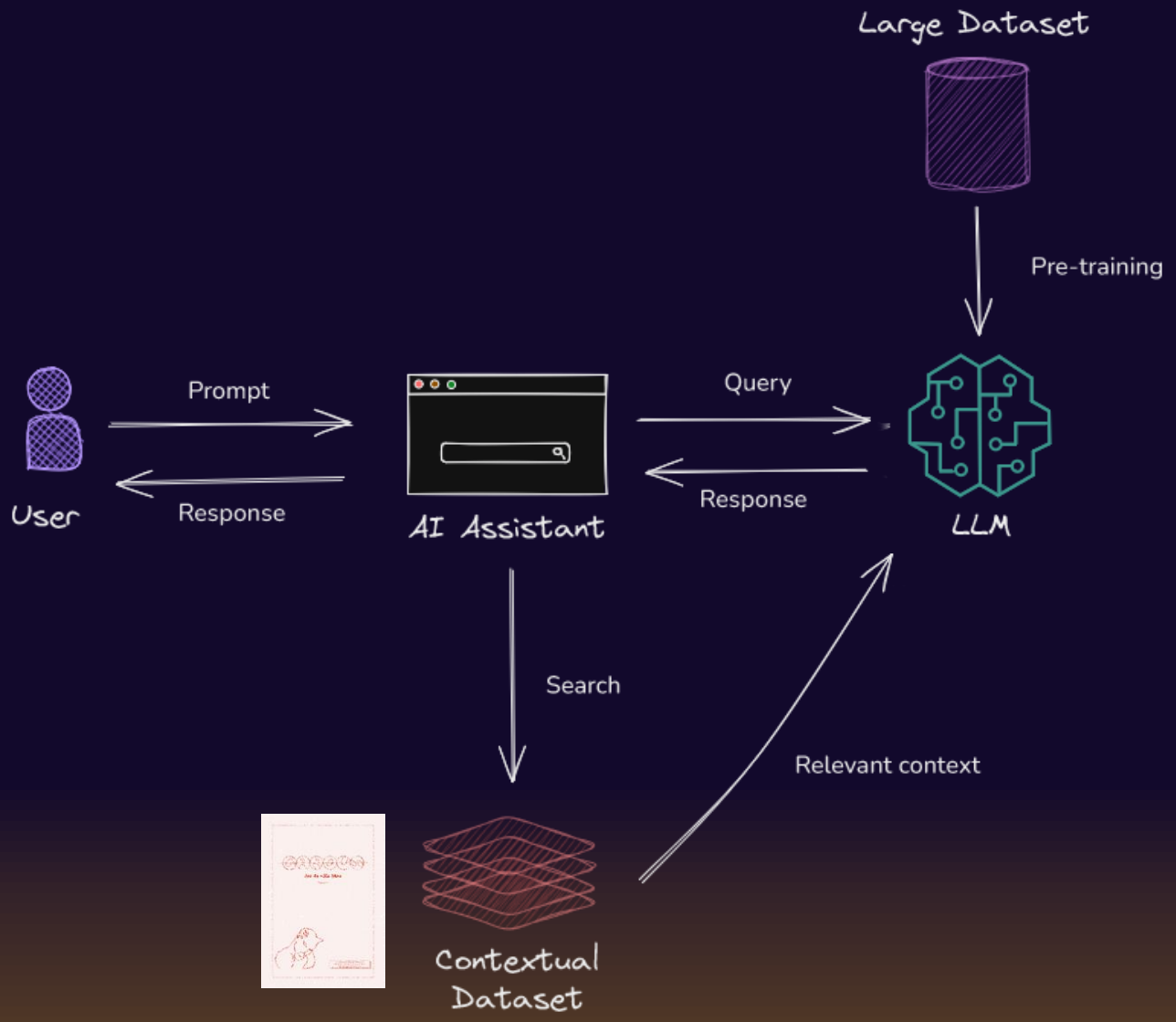
ChatGPT



devday GET INSPIRED

WORLDLINE

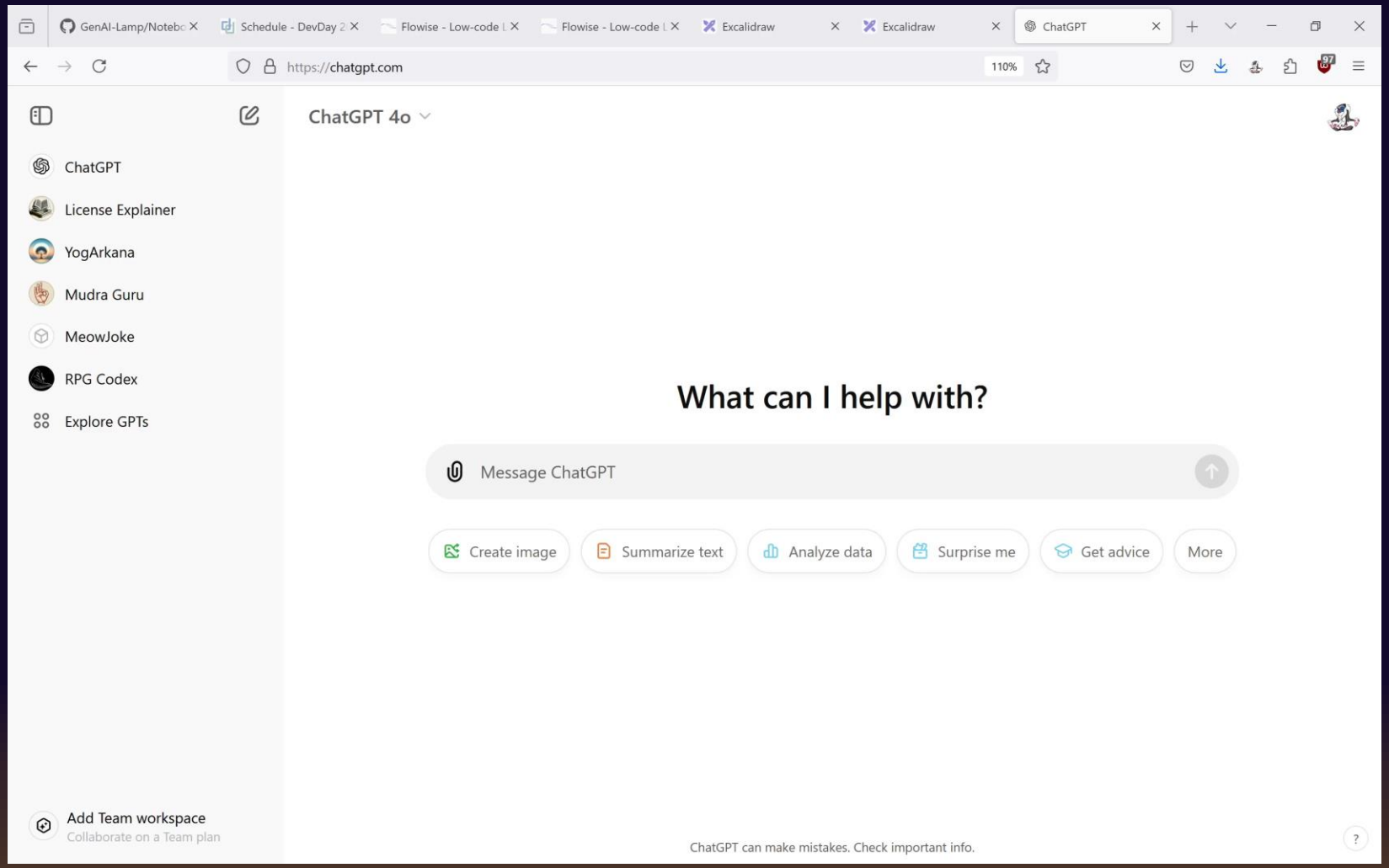
Retrieval Augmented Generation



Retrieval Augmented Generation

DEMO

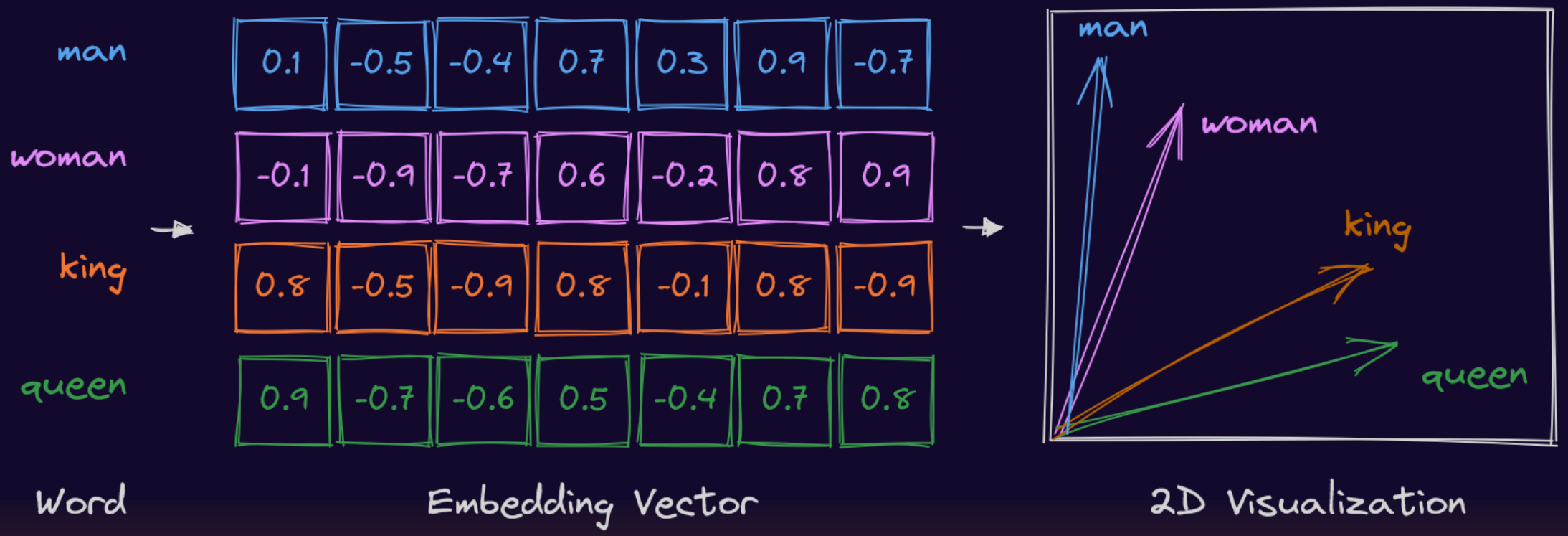
ChatGPT



devday GET INSPIRED

WORLDLINE

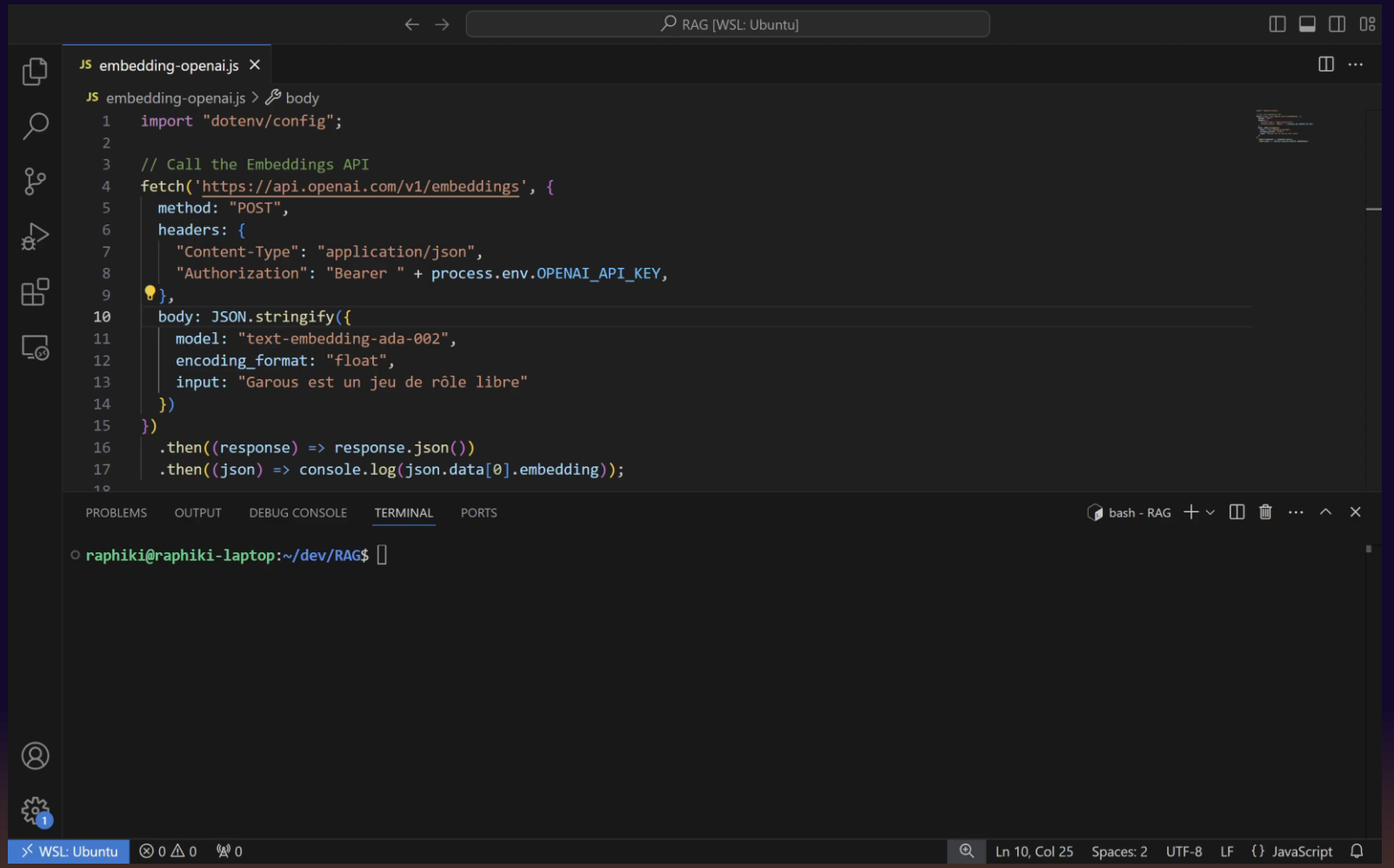
Embeddings Vectors



Embeddings Vectors

DEMO

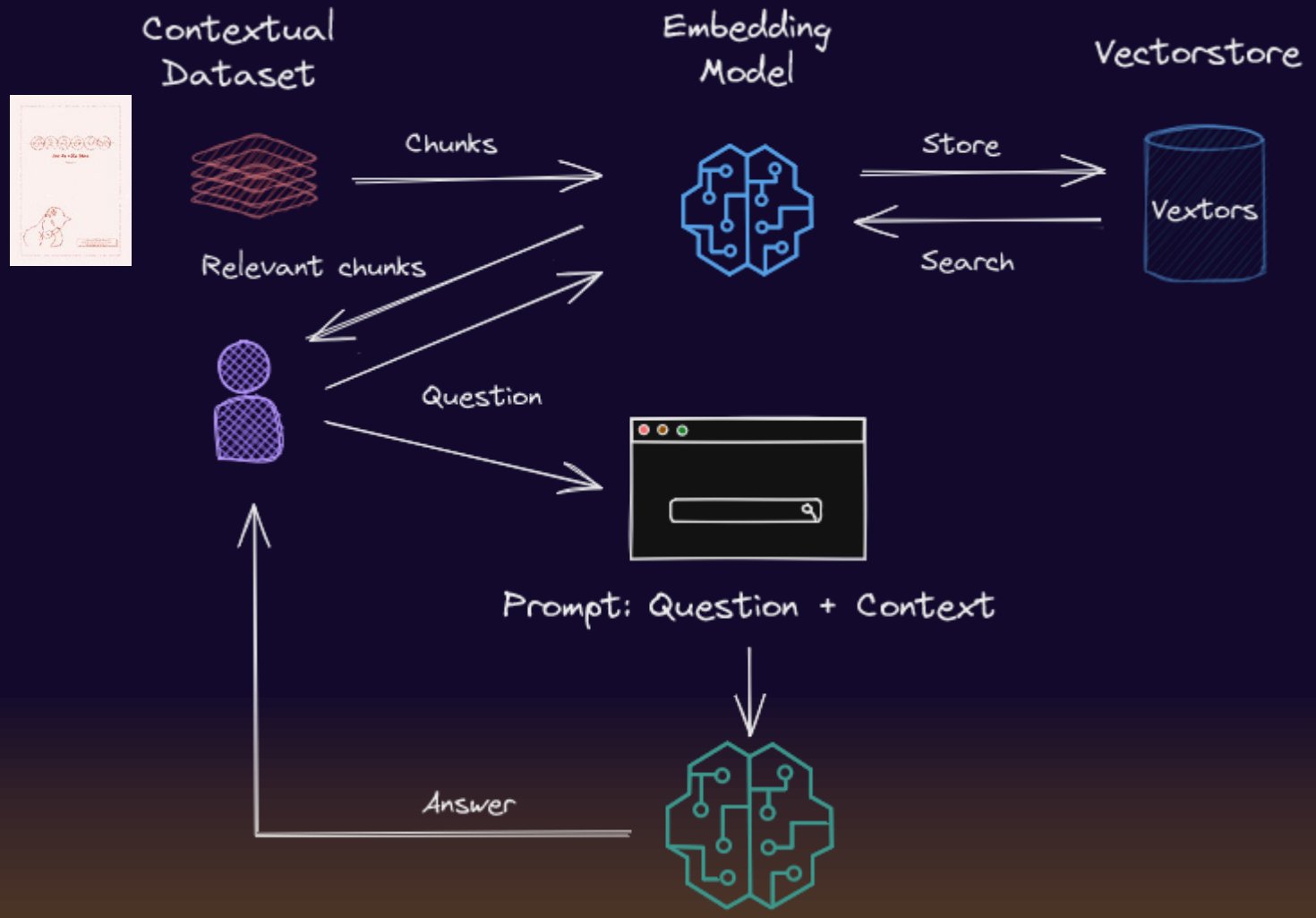
node.js

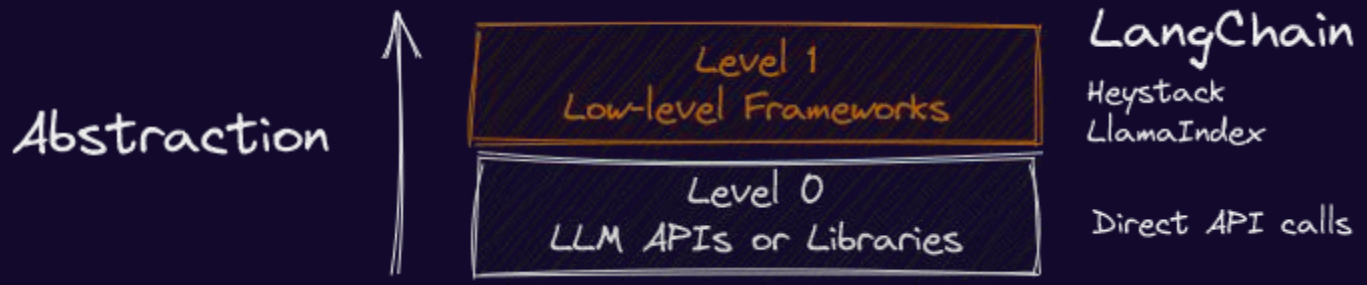


devday GET INSPIRED

WORLDLINE

Embedding & Semantic Search

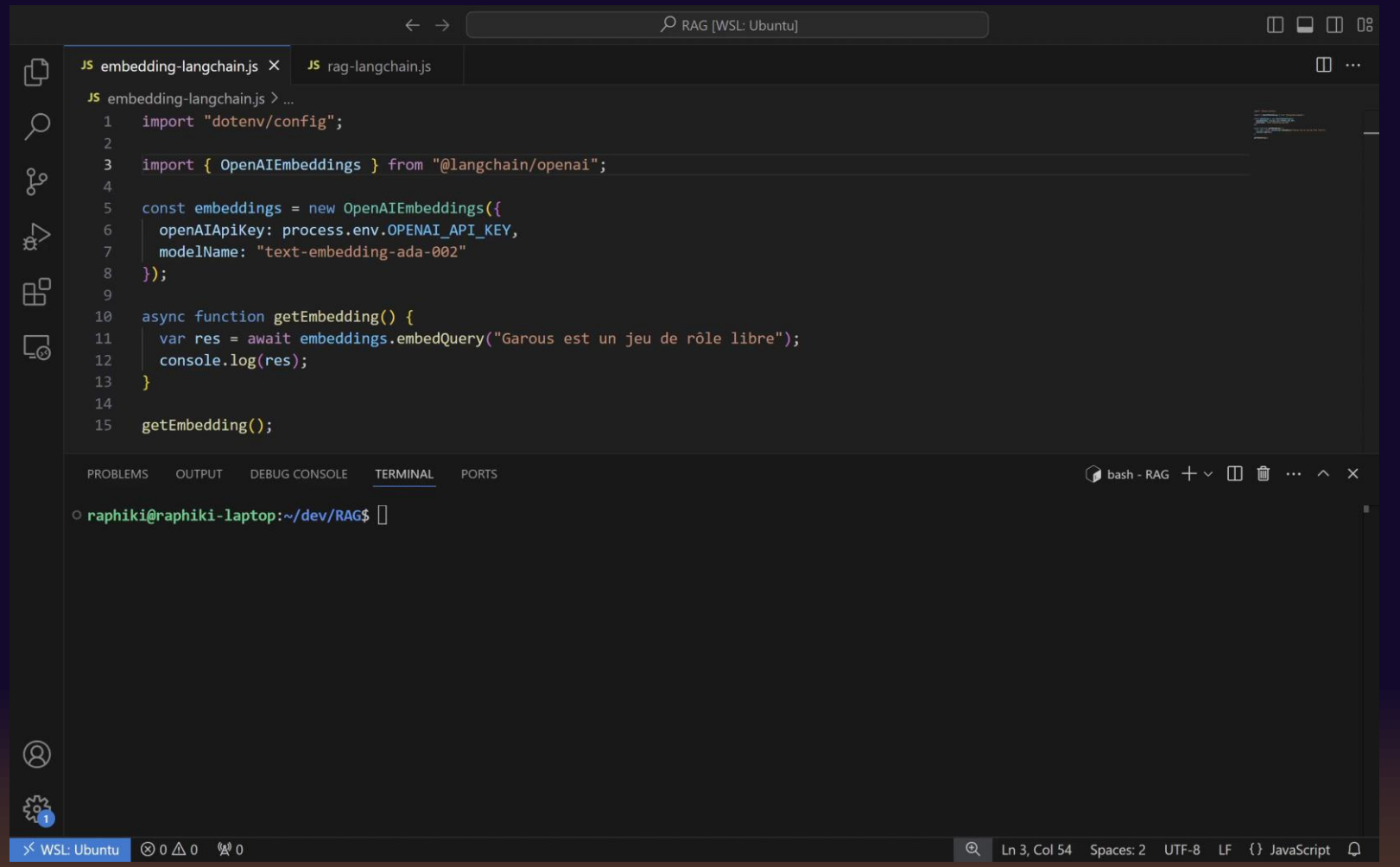



Abstraction
Level 1
Low-level Frameworks
Level 0
LLM APIs or Libraries
LangChain
Heystack
LlamaIndex
Direct API calls

Embedding & Semantic Search

DEMO

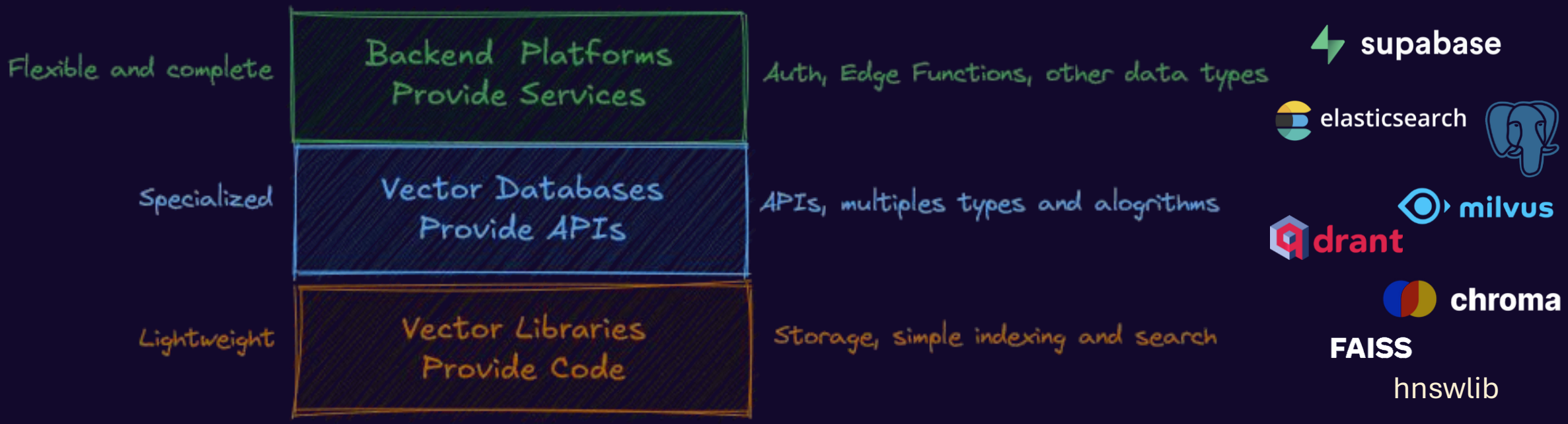
LangChain



devday GET INSPIRED

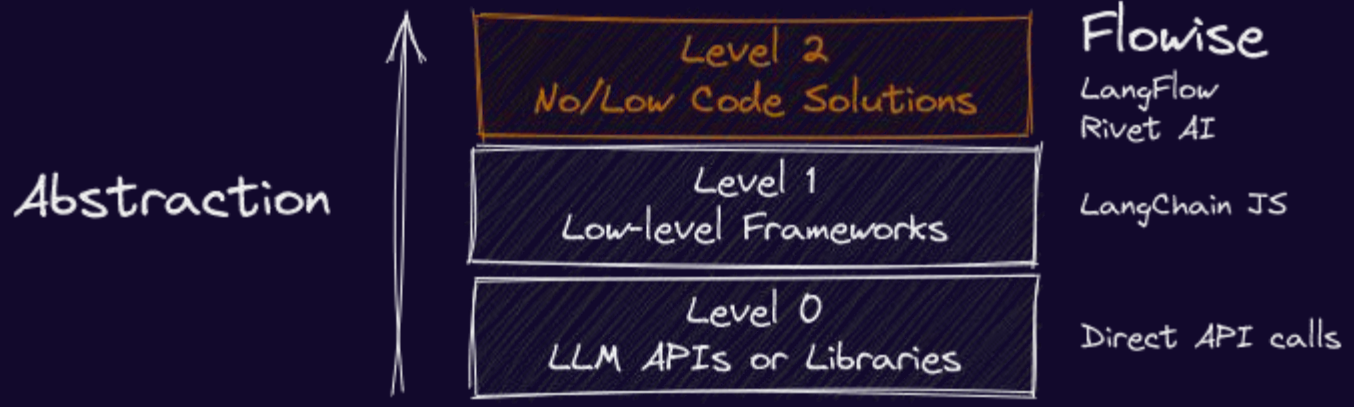
WORLDLINE

Vector Stores

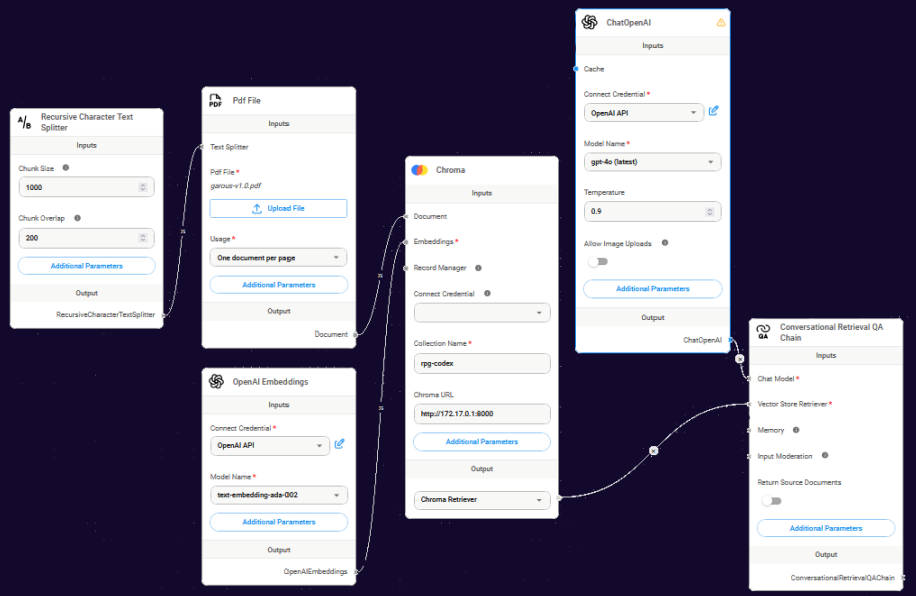


Utilisons Chroma pour notre cas d'usage

chroma


Abstraction
Level 2
No/Low Code Solutions
Level 1
Low-level Frameworks
Level 0
LLM APIs or Libraries
Flowise
LangFlow
Rivet AI
LangChain JS
Direct API calls

Flowise : IHM au-dessus de LangChain JS

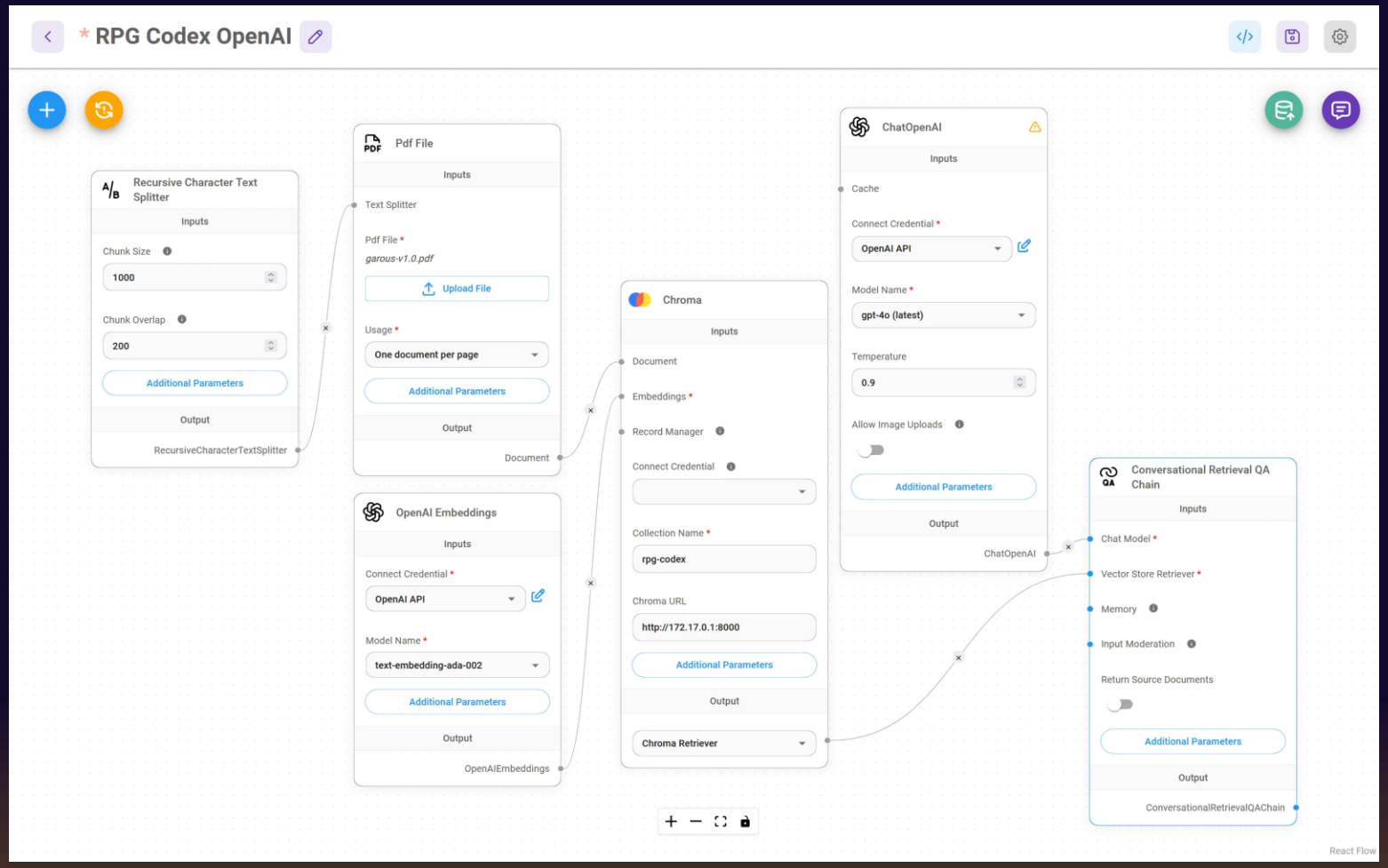


- Solution no-code
- Support des mêmes modèles et vector stores que LangChain JS
- Open Source (license Apache 2)
- Support plus récent de LlamaIndex et d'AgentFlows
- Marketplace de workflows

Flowise : IHM au-dessus de LangChain JS

DEMO

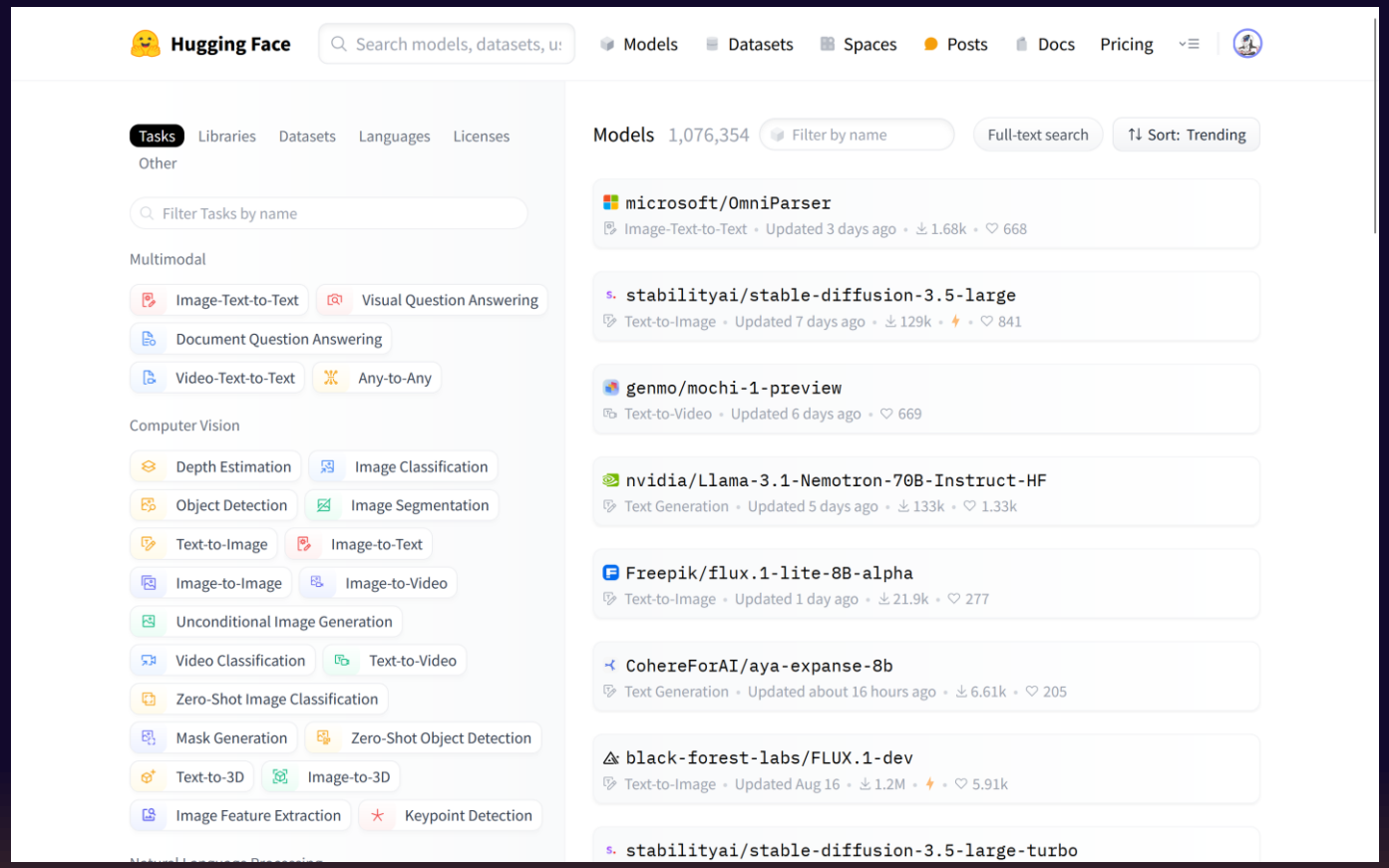
FlowiseAI



devday GET INSPIRED

WORLDLINE

Executer des modèles en local

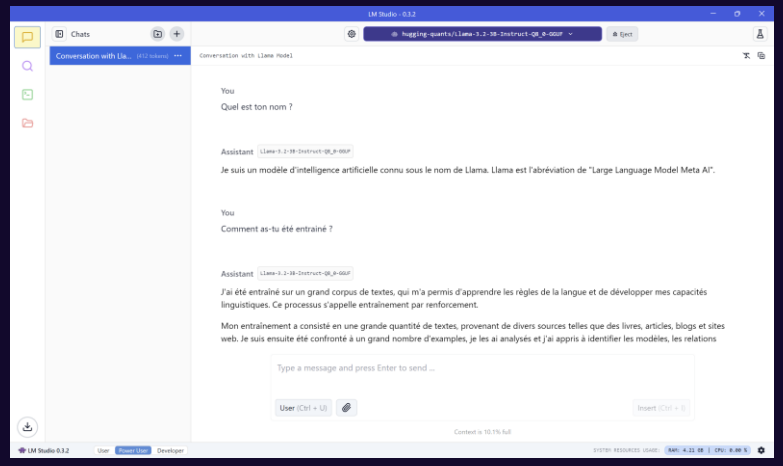


Hugging Face

- Modèles “ouverts”
- Requierent de la RAM
- GPU
- CPU (quantization)

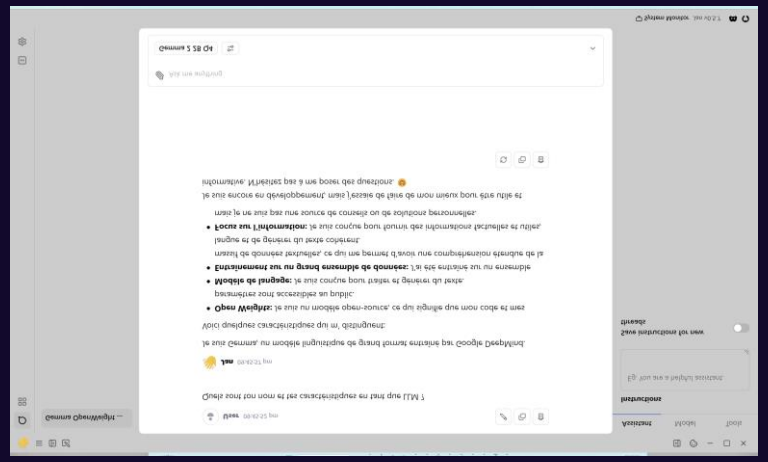
Executer des modèles en local

LM Studio



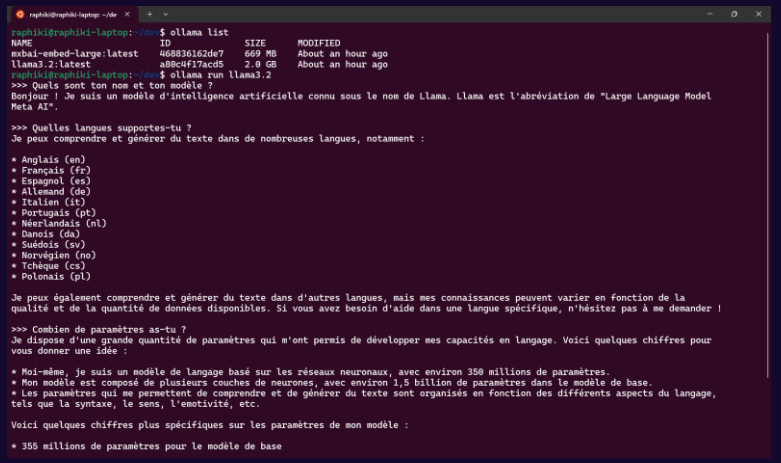
- Pas open source
- IHM
- Mac, Windows, Linux
- Mode server
- Pas d'embeddings

Jan



- Open source
- IHM
- Mac, Windows, Linux
- Mode server
- Pas d'embeddings

Ollama

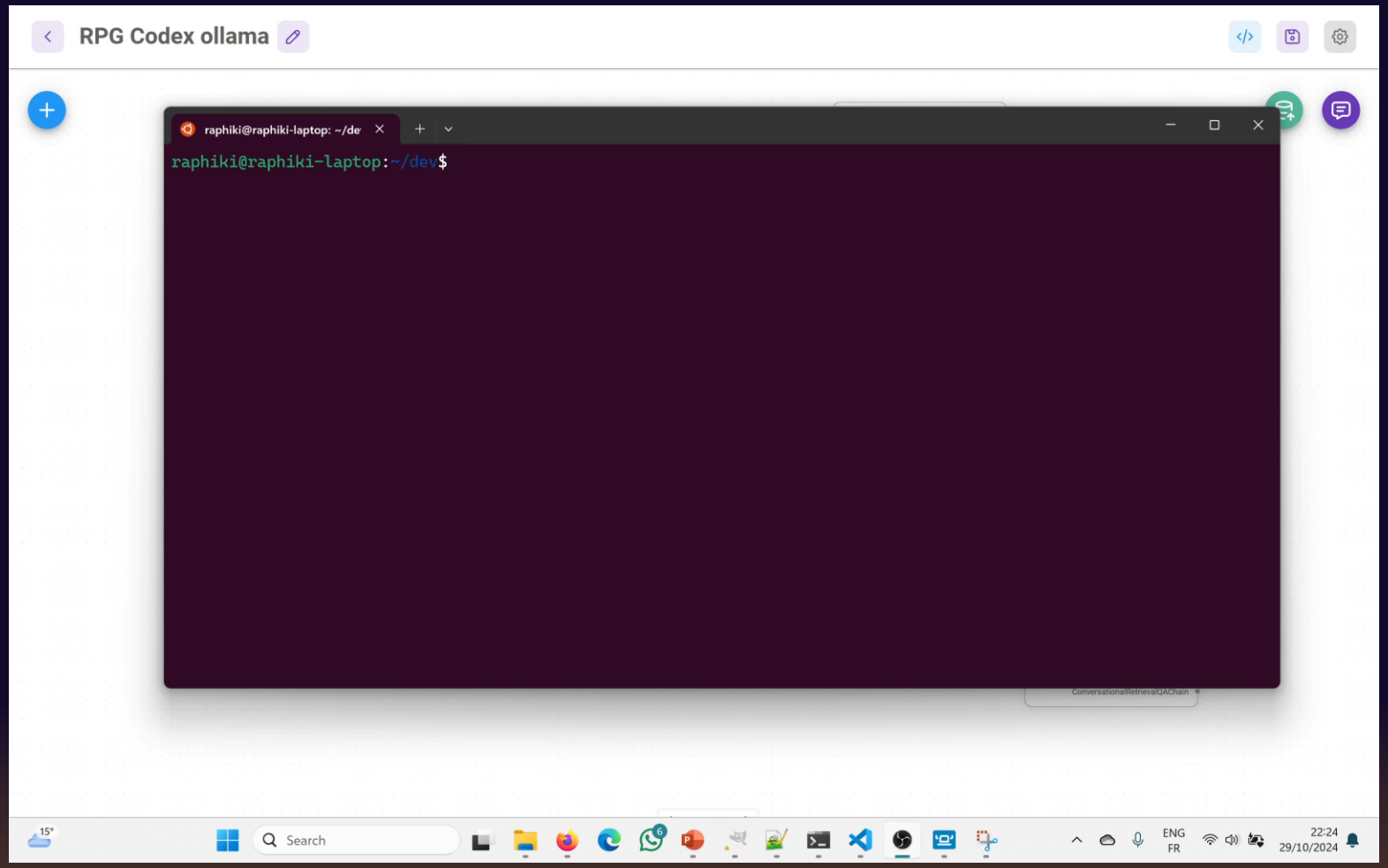


- Open source
- Pas d'IHM
- Mac, Windows, Linux
- Mode server
- Embeddings

Executer des modèles en local

DEMO

Ollama



devday GET INSPIRED

WORLDLINE

Composants ouverts en local

Ollama

chroma

FlowiseAI



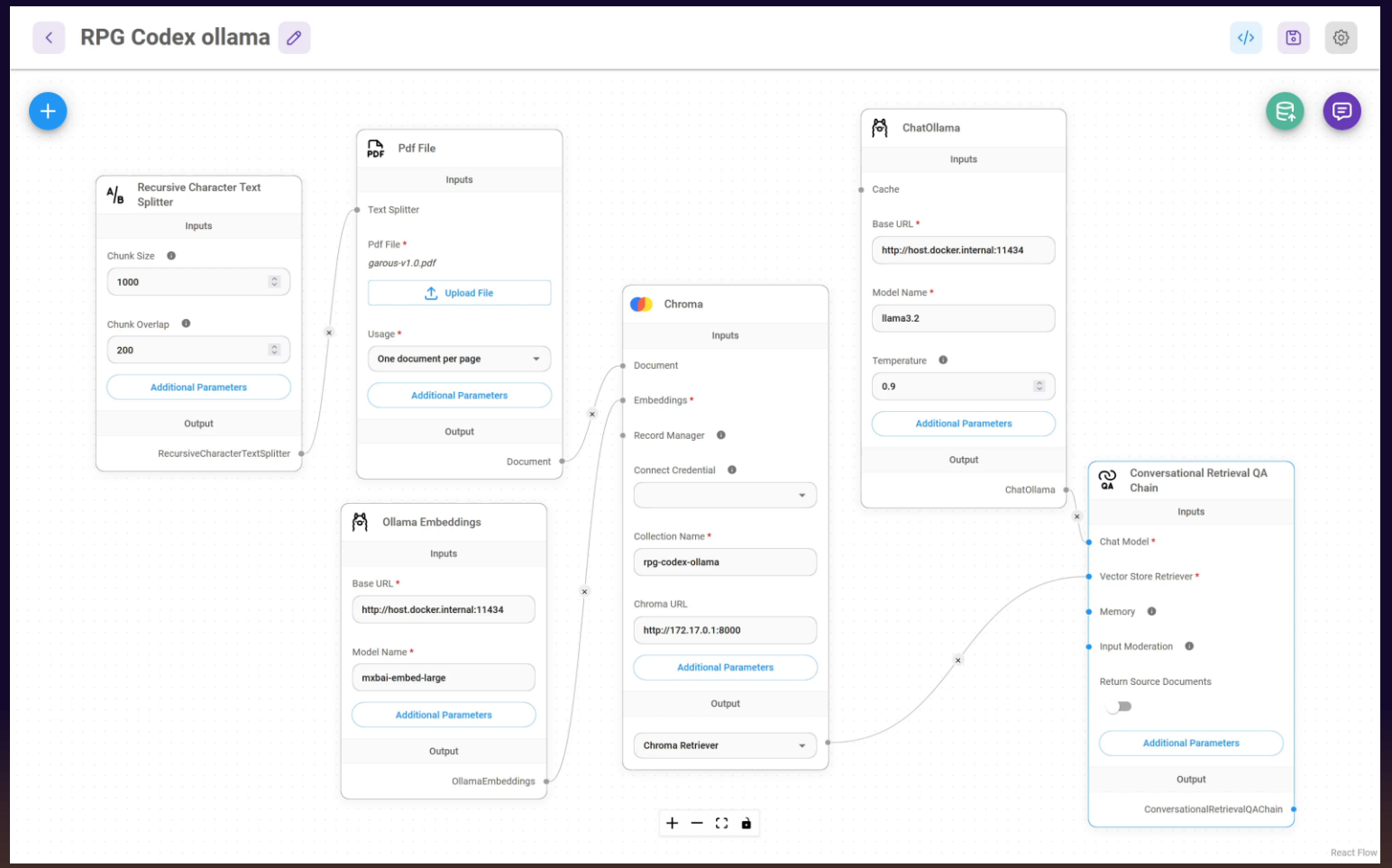
Hugging Face

meta-llama/**Llama-3.2-3B-Instruct**

mixedbread-ai/**mxbai-embed-large**

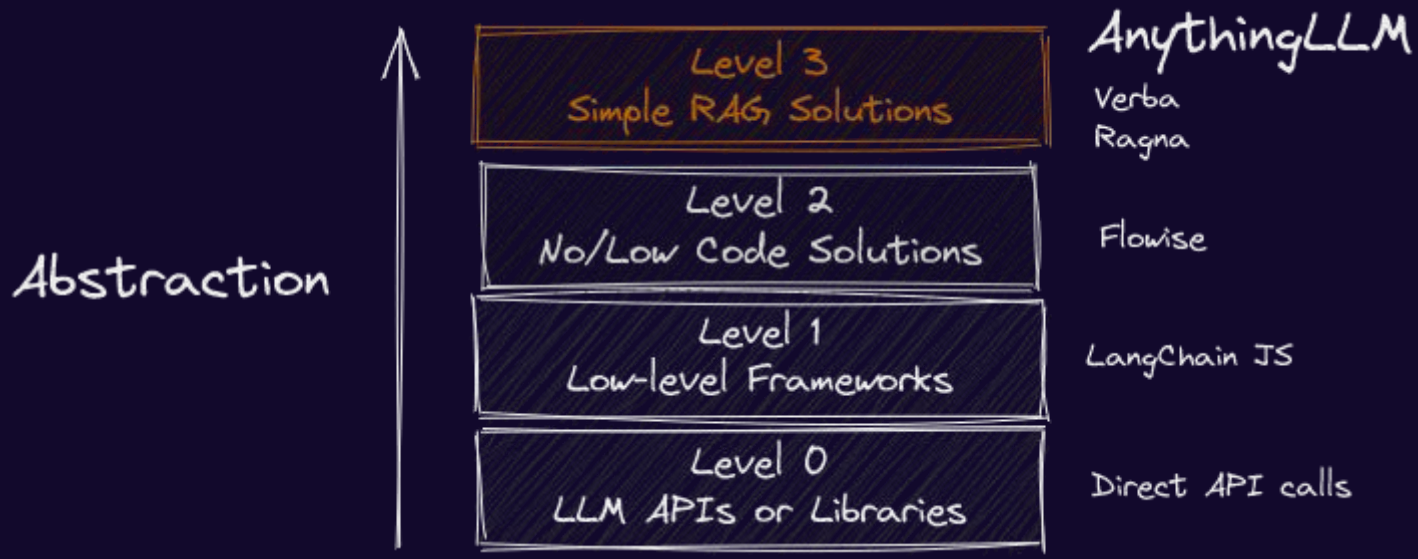
Composants ouverts en local

DEMO
FlowiseAI



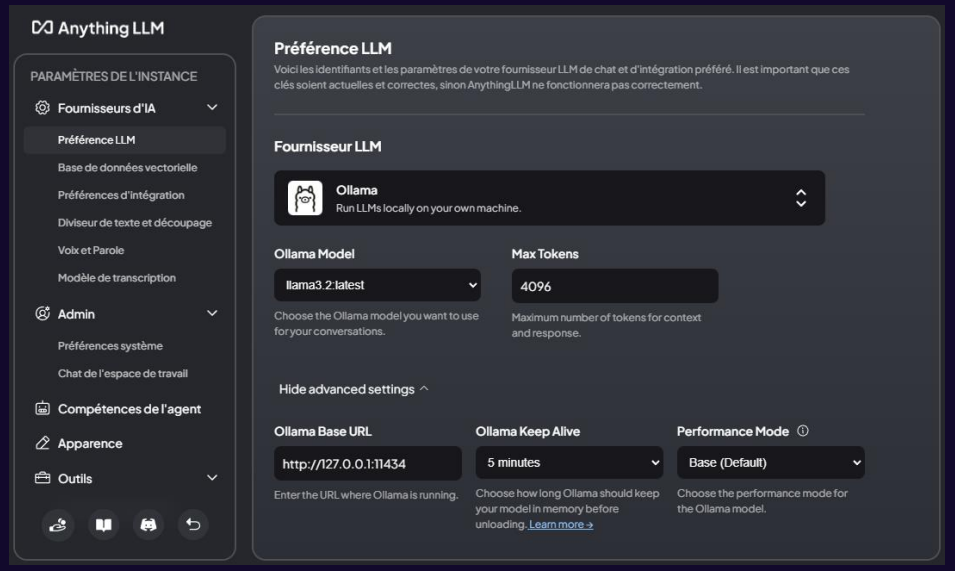
devday GET INSPIRED

WORLDLINE


Abstraction
Level 3
Simple RAG, Solutions
Level 2
No/Low Code Solutions
Level 1
Low-level Frameworks
Level 0
LLM APIs or Libraries
AnythingLLM
Verba
Ragna
Flowise
LangChain JS
Direct API calls

AnythingLLM : solution intégrée

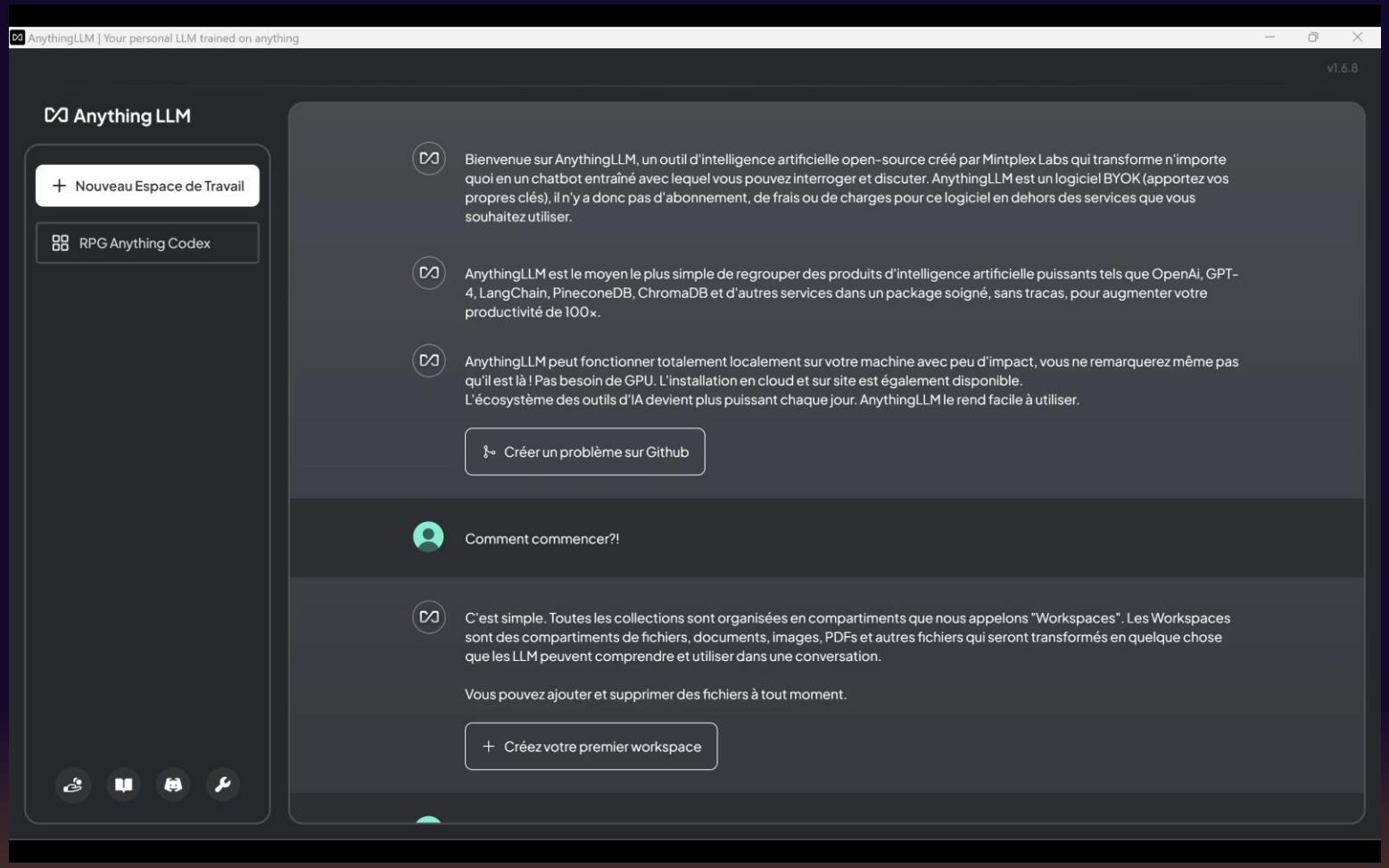
Anything LLM



- Mac/Windows/Linux
- Desktop ou Docker/SaaS (multi-users, chat widget)
- Support de nombreux modèles et vector stores
- Open Source (license MIT)
- RAG, voix (whisper), tools (chart, web search, SQL, custom...)

AnythingLLM : solution intégrée

DEMO
Anything LLM



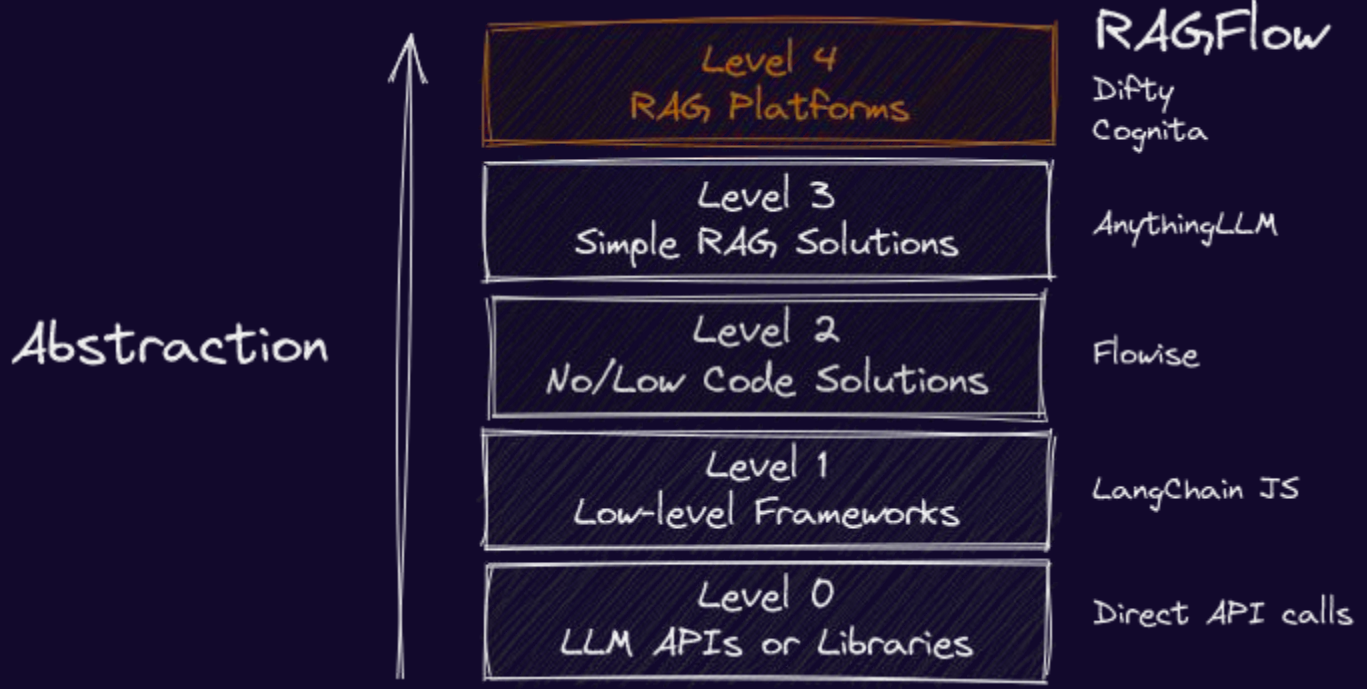
devday GET INSPIRED

WORLDLINE

Techniques de RAG avancées

- **Limitations du Naive RAG**
  - Imprecision, chunks non pertinents
  - Information pertinente manquante
  - Hallucinations, non-alignment

- **Optimisations de l'Advanced RAG**
  - Pre-Retreival : indexation améliorée, métadonnées, re-aligement
  - Retreival : recherche hybride, recursion, sous requêtes
  - Post-Retreival : reranking, document scoring and filtering


Abstraction
Level 4
RAG Platforms
Level 3
Simple RAG Solutions
Level 2
No/Low Code Solutions
Level 1
Low-level Frameworks
Level 0
LLM APIs or Libraries
RAGFlow
Dify
Cognita
AnythingLLM
Flowise
LangChain JS
Direct API calls

RAGFlow : optimiser le RAG

RAGFlow

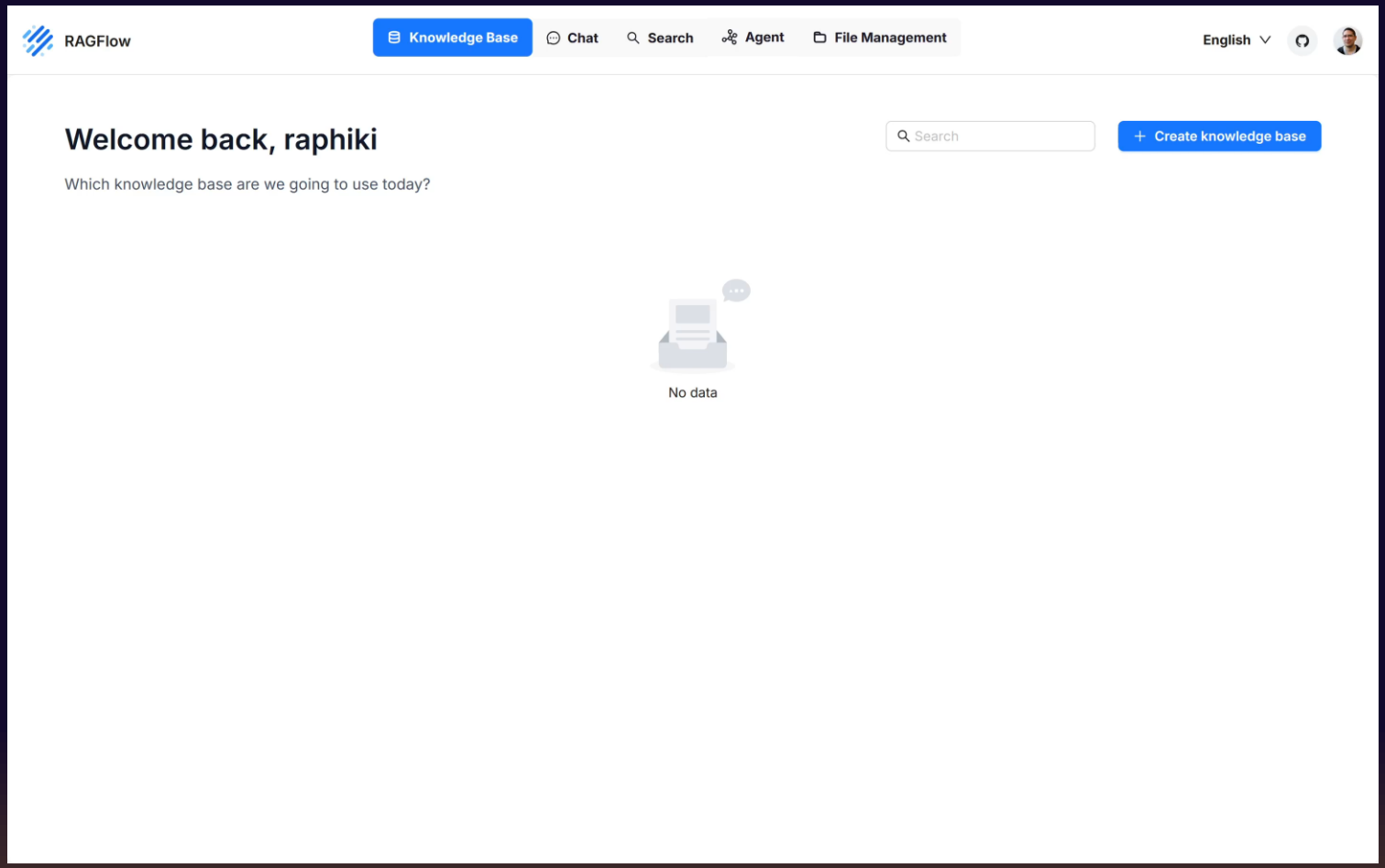


- Architecture multi composants (MySQL, ElasticSearch, Redis, MinIO...)
- Support de nombreux modèles
- Open Source (license Apache 2)
- Certaines techniques de RAG avancé,
- Agents (workflows)

RAGFlow : Knowledge Base

DEMO

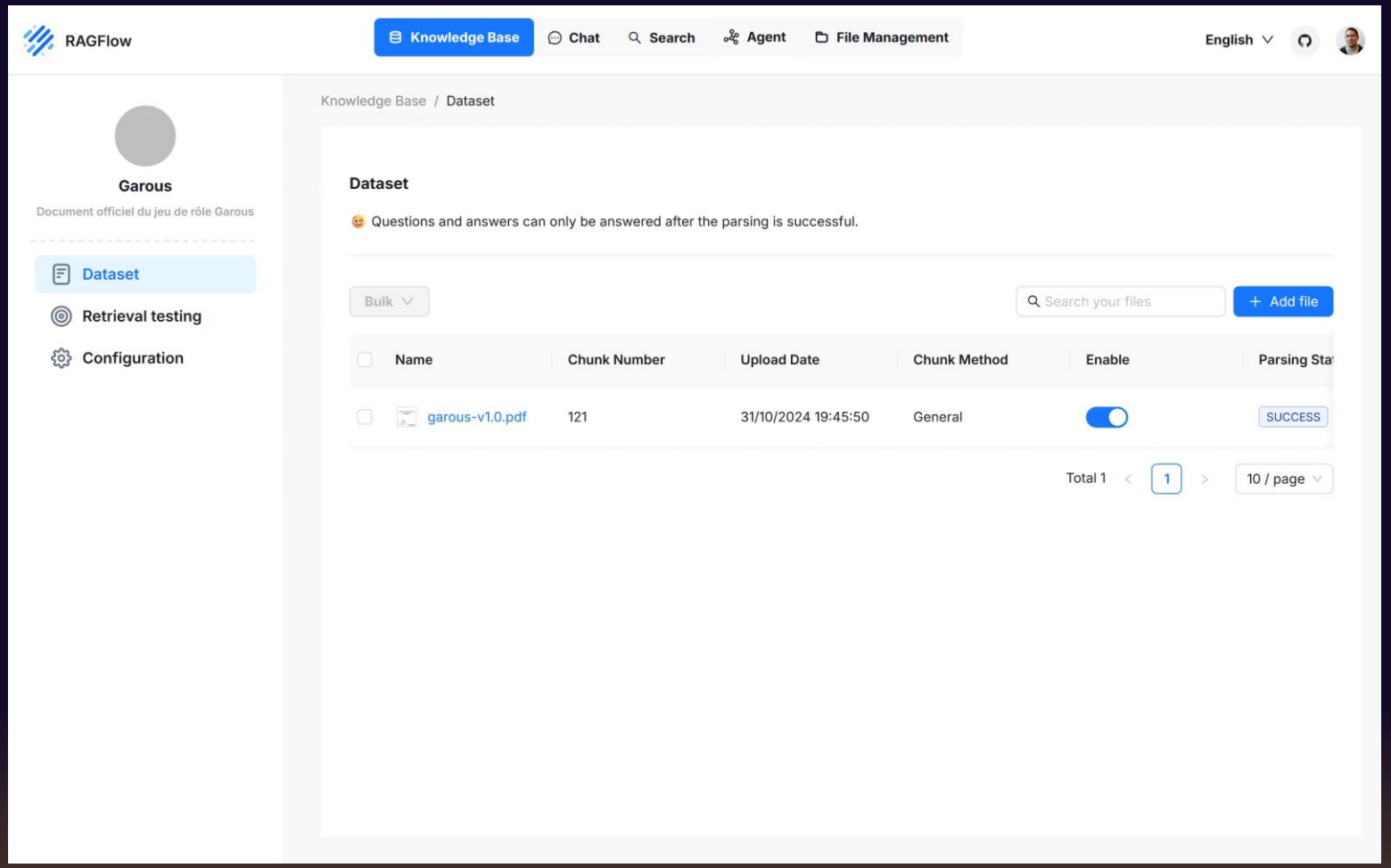
RAGFlow



RAGFlow : Chat avec RAG

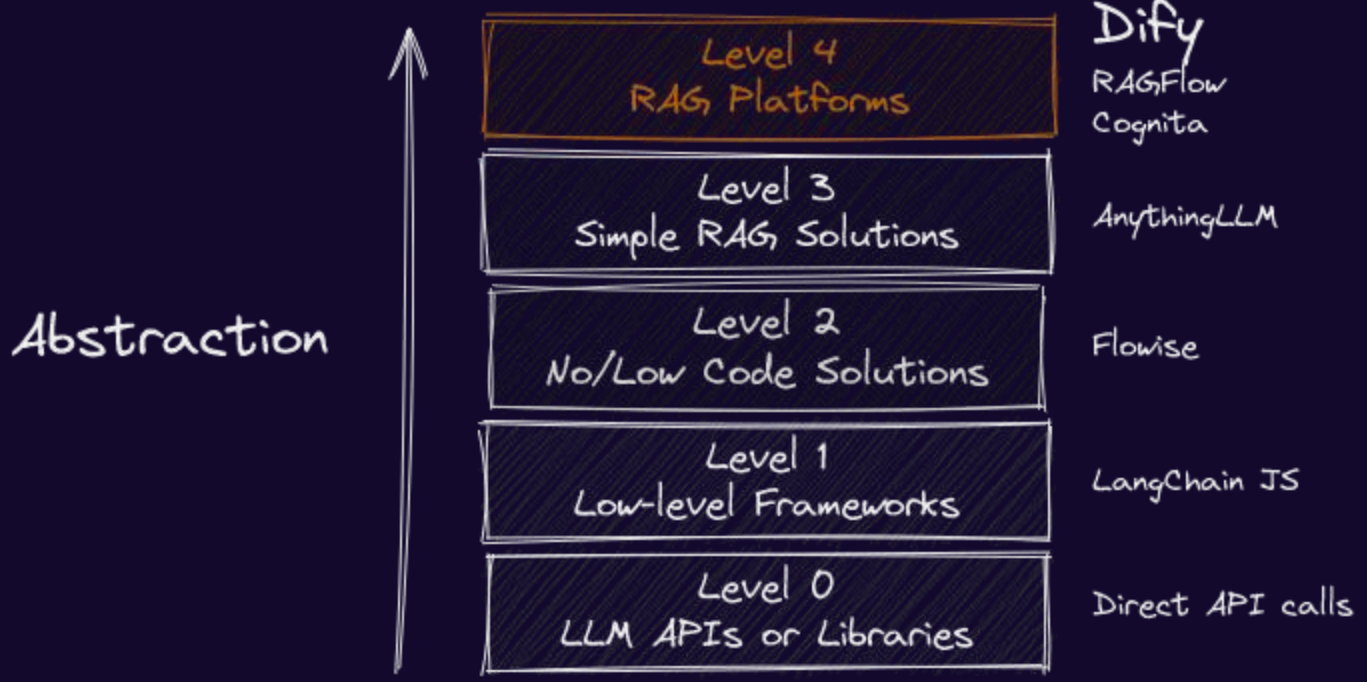
DEMO

RAGFlow

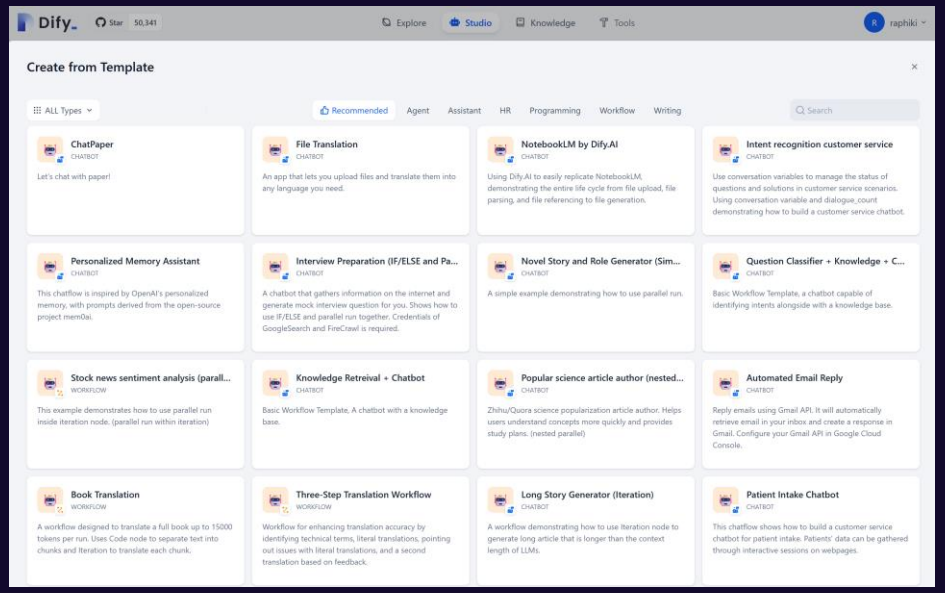


devday GET INSPIRED

WORLDLINE


Abstraction
Level 4
RAG Platforms
Level 3
Simple RAG Solutions
Level 2
No/Low Code Solutions
Level 1
Low-level Frameworks
Level 0
LLM APIs or Libraries
Dify
RAGFlow
Cognita
AnythingLLM
Flowise
LangChain JS
Direct API calls

Dify : générer des assistants et des agents

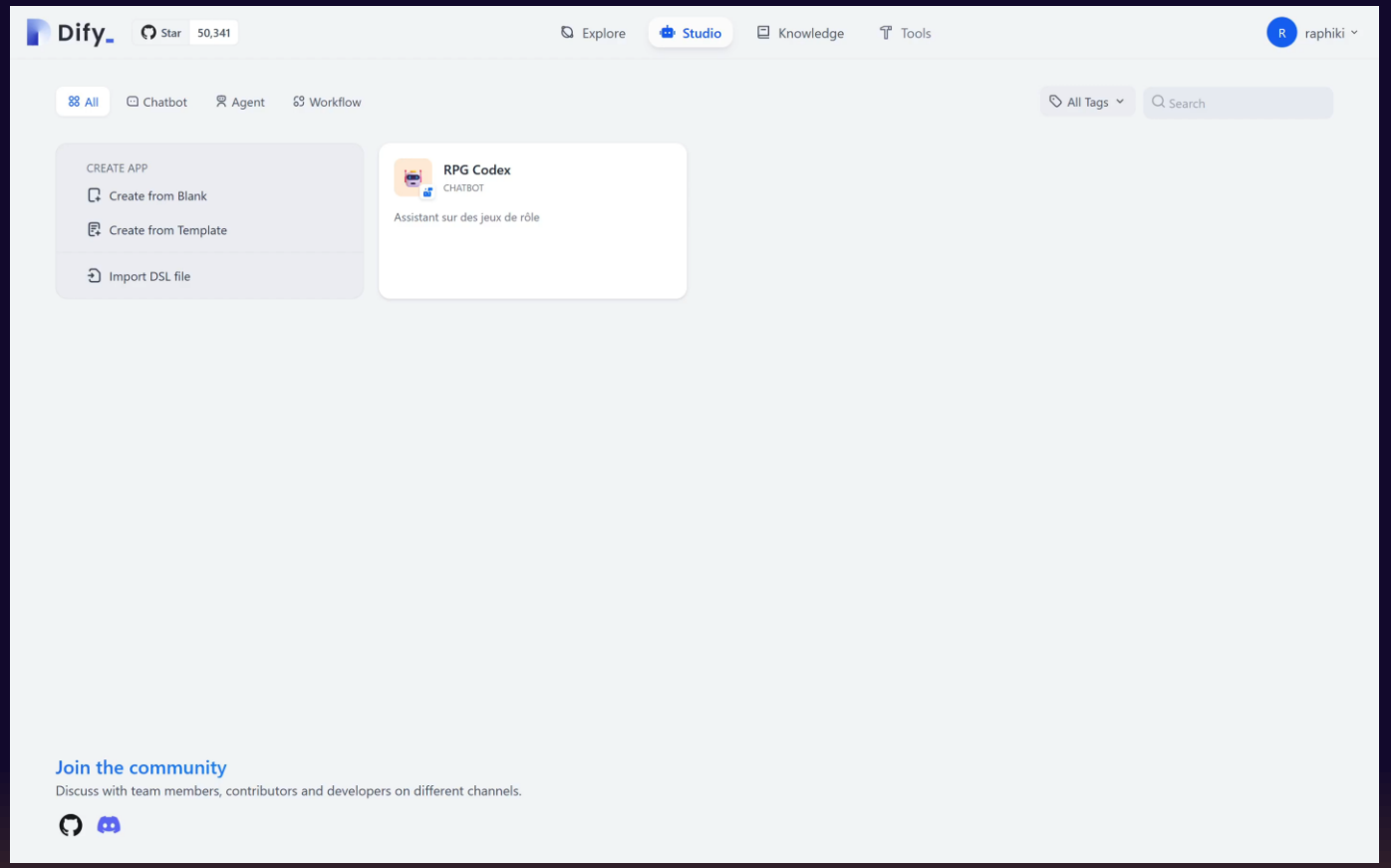


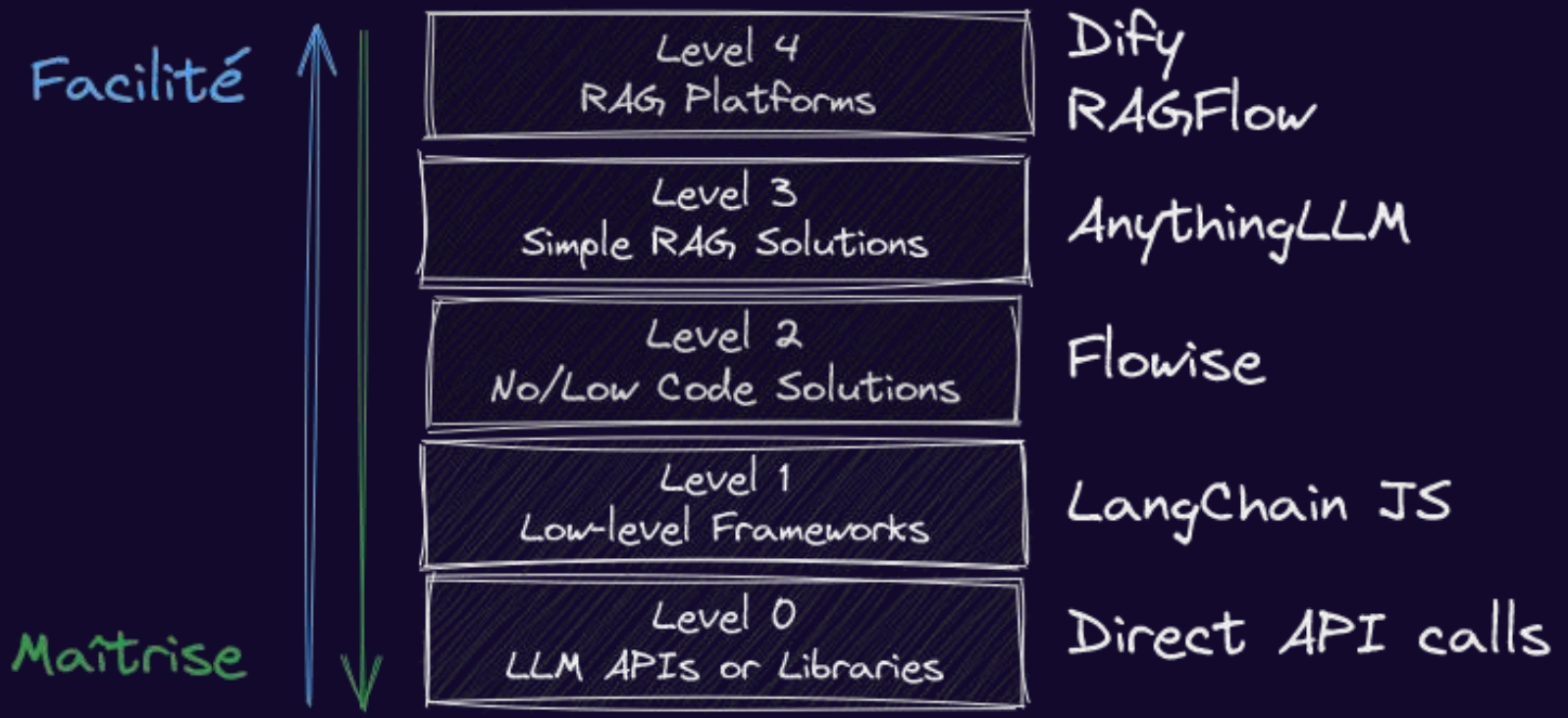
- Architecture multi composants (Postgres, Redis, SSRF proxy, MinIO, vector stores...)
- Support de nombreux modèles
- Open (license Apache 2 modifiée)
- Certaines techniques de RAG avancé
- Notion de workflows et de tools
- Publication des assistants

Dify : générer des assistants et des agents

DEMO

Dify_





Ajuster le niveau d'abstraction en fonction du ratio effort/qualité

De plus en plus de composants open source disponibles

Raphaël **Semeteys**

raphiki.github.io

MERCI !

Feedback



devday GET INSPIRED

WORLDLINE

PLATINIUM

ONIRYX

GOLD

delaware

GAMING1

GENESIS CONSULT
IT Consultancy Services

PARTNER

birdit

DevApps.be

meet innovate create

SENSE OF TECH

Sparkle

devday
GET INSPIRED

WORLDLINE